Low-Temperature Thermal Conductivity and Heat Capacity Measurements of Ordinary and Carbonaceous Chondrites

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Abstract: **We have measured the thermal conductivity and heat capacity at low temperatures (5–300 K) of several meteorites representing a range of compositions, including the ordinary chondrites Gao-Guenie (H4) and Jiddat al Harasis 073 (L6), and the carbonaceous chondrites NWA 6567 (CV3) and NWA 7678 (CV3). These measurements are among the first direct low temperature measurements of thermal conductivity and heat capacity for meteorites. Our results compare well with previous estimates for similar meteorites, where conductivity was derived from diffusivity measurements and modeled heat capacities; our new values are of a higher precision and cover a wider range of temperatures. If the rocky material that makes up asteroids and provides the dust for comets, Kuiper Belt objects, and icy satellites has similar thermal conductivities as the ordinary and carbonaceous chondrites measured here, this data would significantly add to the understanding for models of their thermal evolution.**

(1) Introduction:

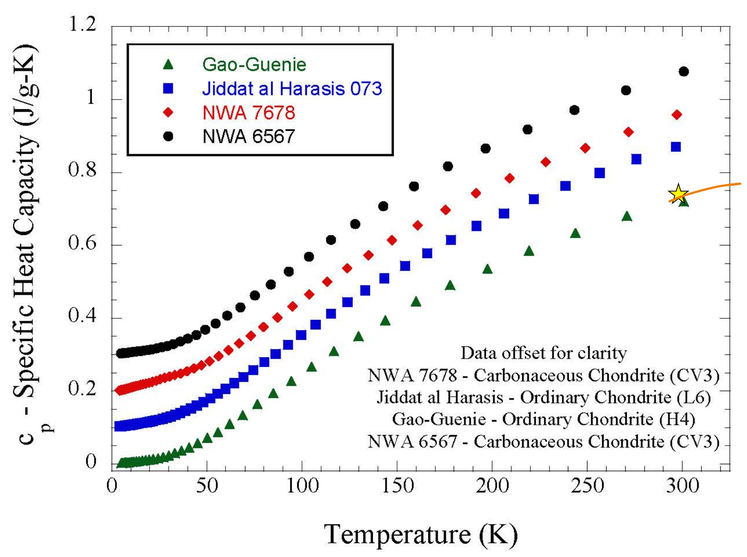
(a) Meteorite heat capacity (specific heat) and thermal conductivity are essential parameters in modeling many aspects of the orbital and internal evolution of small solar system bodies, and can be tools for characterization of the material in a meteorite itself.

(b) While most meteoritic heat capacities are measured at $T \geq 295$ K, our previous work on the shergottite Los Angeles [Opeil, *et al.* (2012)] indicates that heat capacity is a strong function of temperature, especially at the lower temperatures found in the asteroid belt. Low-temperature studies of heat capacity and thermal conductivity may spark further discussion in the astronomy community and an evolution in the theory behind asteroid formation.

(c) These measurements were made on an automated cryogenic He^4 refrigerator, a Physical Properties Measurement System (PPMS) manufactured by Quantum Design. It operates at $2 \leq T \leq 400$ K, $0 \leq H \leq 90{,}000$ Oe and calibrations are referenced to NIST standards.

(2) Specific Heat Capacity vs. T (5-300 K)

Figure 1. Specific heat capacity vs. temperature measurements are shown. The data is offset by +0.1 [J/g-K] for each sample in the vertical direction for clarity. Beech, *et al.*, (2009) report c_p= 732.0 [J/kg-K] at 296.15 K for Gao-Guenie as indicated on the plot by the yellow star. ★ This trend shown for Gao-Guenie is consistent with the curve at higher temperatures reported by Szurgot, *et al.*, (2012) and indicated by the solid orange line.



(3) Modeling Low-Temperature Heat Capacity

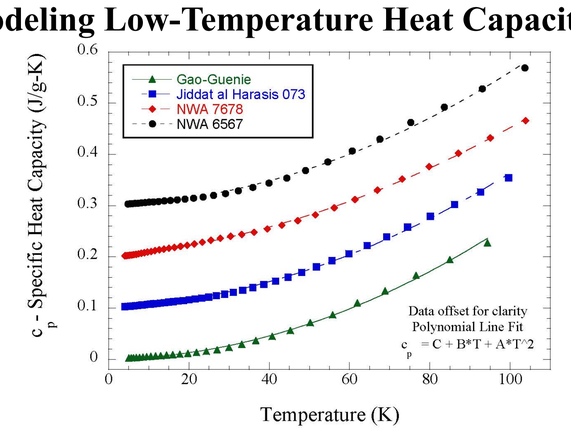


FIGURE 2. The low-temperature specific heat capacity data are fit to a second order polynomial ($c_p = C + B*T + AT^2$) for $T \leq 100$ K. The fitting coefficients A, B, & C are recorded in Table I, below. The data are offset by +0.1 [J/g-K] in the vertical direction for clarity. The lines are added a guide to the eye and represent the polynomial fit to the data $5 \leq T \leq 100$ K.

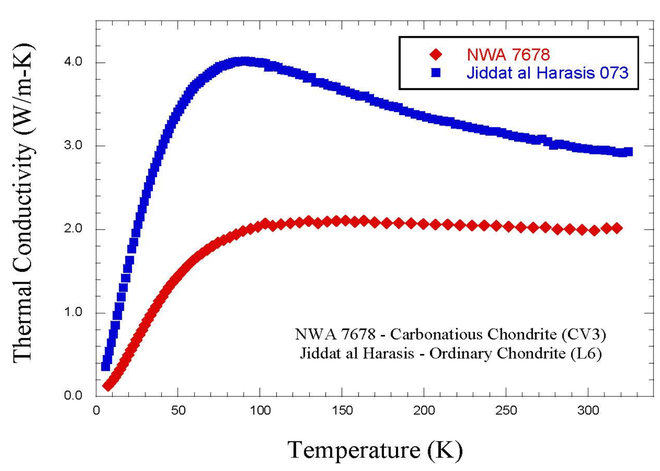
$$c_p = C + B*T + A*T^2$$

Meteorite Name	*A*	*B*	*C*	*Poly-fit*	*ρ (g/cm³)*	*M_{HC} (mg)*	*M_{TC} (g)*
Gao-Guenie	0.00002522	0.00012323	0.00031143	0.99880	3.51 (0.03)	17.8	N/A
Jiddat al Harasis	0.00002275	0.00035745	0.00076123	0.99939	3.29 (0.03)	24.7	0.7991
NWA 7678	0.00001788	0.00073536	0.00005616	0.99963	2.99 (0.03)	26.2	0.7025
NWA 6567	0.00002348	0.00025934	0.00028777	0.99866	3.30 (0.03)	28.4	0.4848

TABLE I: Sample dimensions for heat capacity and thermal conductivity were approximately 1 x 2 x 3 mm and 3 x 3 x 14 mm respectively. Our density measurements compare favorably with ref. Consolmagno, *et al.* (2006), Consolmagno, *et al.*, (2013), and Gnos, *et al.*, (2009).

(4) Thermal Conductivity vs. T (5-300 K)

FIGURE 3. Thermal conductivity vs. temperature data for NWA 7678 and Jiddat al Harasis 073 are shown. The thermal conductivity sample dimensions were approximately 3 x 3 x 14 mm. A cooling rate was set at 0.5 K/min. These data are consistent our with other thermal conductivity measurements made with samples cut from different spatial orientations in the same parent meteorite.



(5) Thermal Diffusivity vs. T calculation

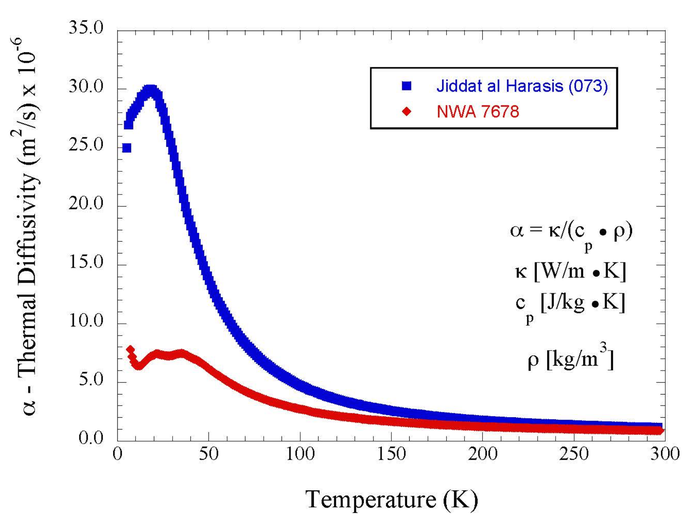


FIGURE 4. Using the measurements of mass density, ρ_{mass}, thermal conductivity, κ, and specific heat capacity, c_p, the thermal diffusivity α, is calculated using the relationship $\alpha = \kappa/(c_p \bullet \rho)$. For this calculation we use the approximation $c_V \sim c_p$ [ref. Scharli and Rybach, (2001)] for NWA 7678 and Jiddat al Harasis (073) over this temperature range.

(6) Conclusions/Analysis:

(a) Because the temperature dependence of specific heat capacity has a considerable effect on thermal models, a greater database of experimental data on specific heat capacity of various mineral compositions and their dependence on temperature is needed, see ref. Ghosh and McSween (1990).

(b) Our preliminary results for ordinary and carbonaceous chondrites are in excellent agreement with the literature data for meteorites, and hold the promise that such measurements may not only produce values useful to modelers, but they also may provide an efficient way to classify whole meteorite samples and characterize subtle differences between meteorites of different compositional classes.

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